Similarity Methods in Navier-Stokes Problems

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Similarity Methods

feature article

Dynamic similarity, the dimensionless science

Diogo Bolster, Robert E. Hershberger, and Russell J. Donnelly

Dimensional analysis, a framework for drawing physical parallels between systems of disparate scale, affords key insights into natural phenomena too expansive and too energetic to replicate in the lab.

Diogo Bolster is an assistant professor of civil engineering and geological sciences at the University of Notre Dame in Notre Dame, Indiana. **Robert Hershberger** is a research assistant in the department of physics at the University of Oregon in Eugene. **Russell Donnelly** is a professor of physics at the University of Oregon.

Overview

1. Rudiments
2. Books
3. Bibliography

Physics Today

feature article

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Bolster, Hershberger, and Donnelly 2011

Buckingham's Pi Theory

Physical laws are independent of the form of the units

Dimensional Analysis and the Pi Theorem

ABSTRACT

A new version of the Buckingham pi theorem is presented which reveals the underlying mathematical structure of that classical result. In this context it becomes a theorem in linear algebra, and it is formulated without reference to physical quantities, units, dimensions, and so on. Also, the classical approach of Birkhoff is reviewed and some points in his proof are expanded.

Curtis, Logan, and Parker 1982

Dimensional Analysis and the Pi Theorem

The most fundamental result in dimensional analysis is the pi theorem.

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One of the most famous examples of dimensional reasoning was the derivation, by G. I. Taylor, of the formula

$$r = t^{2/5} \left(\frac{E}{\rho_0} \right)^{1/5} f(\gamma) \tag{1.1}$$

Dimensional Analysis and the Pi Theorem

Assumption: $\exists$ physical law of the form

$$g(t, T, \rho_0, E, \gamma) = 0 \tag{1.2}$$

Dimensional Analysis and the Pi Theorem

Two dimensionless quantities

$$\Pi_1 = \gamma, \quad \Pi_2 = \frac{r^5 \rho_0}{t^2 E} \tag{1.3}$$

Dimensional Analysis and the Pi Theorem

Definition (Physical Law)

A physical law $f(Q_i, ..., Q_m) = 0$ relates dimensional quantities $Q_i, ..., Q_m$.

Dimensional Analysis and the Pi Theorem

Theorem (Pi Theorem)

Let L be a law on V compatible with T. Then there exist frames e such that $T(e_k) = 0$ for $k = r + 1, \ldots m$,

Proof.

□

Similarity

1. Nuclear Explosions (Taylor 1950a; Taylor 1950b)
2. Astrophysics (Ostriker and McKee 1988)
3. Laser driven (Edwards et al. 2001)
4. Plasma (Moore, Symes, and Smith 2005)
5. Granular (Boudet and Kellay 2013; Boudet, Cassagne, and Kellay 2009)

Sedov's Formal Development

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Equations of Motion

148 SIMILARITY AND DIMENSIONAL METHODS IN MECHANICS

or chemical change (we shall consider later the extent to which these assumptions are valid in any given problem). The equations of motion, continuity and energy take the form

$$\left.\begin{aligned} \frac{\partial v}{\partial t}+v\frac{\partial v}{\partial r}+\frac{1}{\rho}\frac{\partial p}{\partial r} &= 0, \\ \frac{\partial \rho}{\partial t}+\frac{\partial \rho v}{\partial r}+(\nu-1)\frac{\rho v}{r} &= 0, \\ \frac{\partial}{\partial t}\left(\frac{p}{\rho^{\gamma}}\right)+v\frac{\partial}{\partial r}\left(\frac{p}{\rho^{\gamma}}\right) &= 0, \end{aligned}\right\} \quad (1.3)$$

where γ is the adiabatic index; $\nu = 1$ for plane flow, $\nu = 2$ for flow with cylindrical symmetry and $\nu = 3$ for flow with spherical symmetry.

Sedov 1993

Units: Familiar Quantities

1. velocity
2. momentum
3. kinetic energy
4. potential energy (gravitational)
5. force
6. pressure
7. density (volumetric)

Units: Familiar Quantities

velocity	v	LT^{-1}
momentum	$p = mv$	MLT^{-1}
kinetic energy	$K = \frac{1}{2}mv^2$	ML^2T^{-2}
potential energy	$U = mgh$	ML^2T^{-2}
force	$F = ma$	MLT^{-2}
pressure	$P = F/A$	MT^{-2}
density	δ	ML^3

Input Parameters with Dimension

1. E, energy
2. ρ_0, air pressure

Rankine-Hugoniot Jump Conditions

Connects physical states across shock front

Rankine-Hugoniot Jump Conditions

Conservation Laws:

mass	$\rho_1 u_1$	$=$	$\rho_2 u_2$
momentum	$\rho_1 u_1^2 + p_1$	$=$	$\rho_2 u_2^2 + p_2$
energy	$h_1 + \frac{1}{2} u_1^2$	$=$	$h_2 + \frac{1}{2} u_2^2$

Rankine-Hugoniot Jump Conditions

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Not conserved: Entropy

Rankine-Hugoniot Jump Conditions

Conservation Laws:

mass	$\rho_1 u_1$	$=$	$\rho_2 u_2$	$\Rightarrow$	u_k	$=$	$\frac{m}{\rho_k}$
mom.	$\rho_1 u_1^2 + p_1$	$=$	$\rho_2 u_2^2 + p_2$	$\Rightarrow$	$p_2 - p_1$	$=$	$m\left(u_1 - u_2\right)$
energy	$h_1 + \frac{1}{2}u_1^2$	$=$	$h_2 + \frac{1}{2}u_2^2$	$\Rightarrow$	$h_2 - h_1$	$=$	$\frac{1}{2}\left(u_1^2 - u_2^2\right)$

Rankine-Hugoniot Jump Conditions

Implications:

$$-m^2 = \frac{p_2 - p_1}{1/\rho_2 - 1/\rho_1} \tag{1.4}$$

$$h_2 - h_1 = \frac{1}{2}\left(1/\rho_2 - 1/\rho_1\right)\left(p_2 - p_1\right) \tag{1.5}$$

Jump Conditions: Physical Properties

Table: default

u_l	u_r	**density**
ρ_l	ρ_r	**velocity**
T_l	T_r	**pressure**

Physical Properties

Table: default

u_l	u_r	**density**
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T_l	T_r	**pressure**

Assumptions, Simplifications

1. point explosion
2. infinitesimal time
3. high energy explosion
4. infinitely strong shock
5.

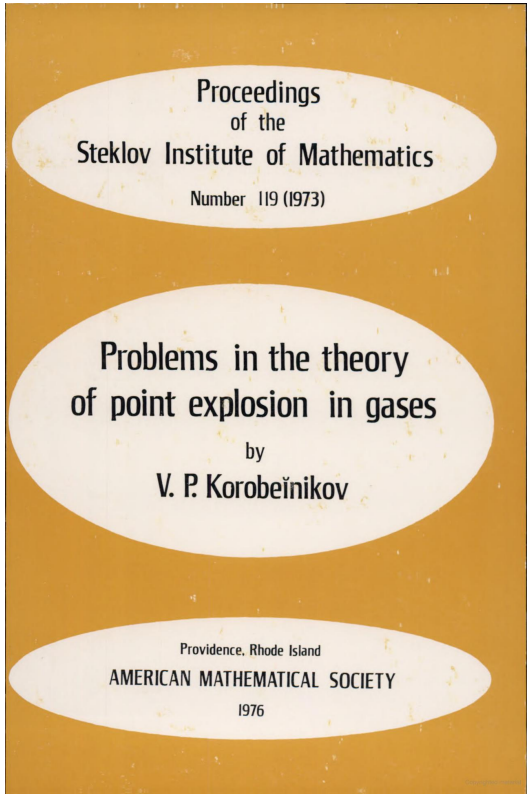
Assumptions, Simplifications

1. spherically symmetric blast wave
2. point source
3. infinitesimal time
4. infinitely strong shock
5.

Self-Similar Region

1. pressure behind shock wave dominates air pressure $\rho_1 >> \rho_0$
2. point source
3. infinitesimal time
4. infinitely strong shock
5.

Problems in the Theory of Point Explosion in Gases



Korobeĭnikov 1976

Invariant Solutions: Cataloged

One-dimensional motions with spherical or cylindrical symmetry have only the invariant solutions:

$$
\begin{array}{llll}
v = r & \rho = e^{\beta t} R(r) & p = e^{\beta t} P(r) & \\
v = r + e^{t} u(\lambda) & \rho = e^{(\beta-2)t} R(\lambda) & p = e^{\beta t} P(\lambda) & \lambda = r e^{-t} \\
v = \frac{1}{t} u(r) & \rho = e^{2(\beta+1)} R(\lambda) & p = t^{2^{\beta}} P(r) & \\
v = \frac{r}{t} u(\lambda) & \rho = e^{(\beta+2\alpha-2)t} R(\lambda) & p = r^{\beta} P(\lambda) & \lambda = t r^{-\alpha}
\end{array}
\tag{1.1}
$$

Korobeĭnikov 1976, p. 51, 1.134–137
Ovsjannikov 1962

Sedov Solutions

Sedov's class of self-similar solutions of 1D gas dynamics

$$
\begin{aligned}
v &= \frac{r}{t}V(\lambda),\\
\rho &= \frac{a}{r^{k+3}t^{s}}R(\lambda),\\
p &= \frac{a}{r^{k+1}t^{s+2}}P(\lambda),\\
\lambda &= \frac{r}{bt^{d}}
\end{aligned}
\tag{1.2}
$$

Dimensional constants: a, b,
Numbers: k, s, δ

Invariant Solutions of Gas Dynamics

Theorem (Invariant Solutions for Gas Dynamics)

All the invariant solutions of the equations of gas dynamics for spherical and cylindrical symmetry are either self-similar solutions or limits of self-similar solutions

Proof.

See Korobeĭnikov 1976, p. 52 □

Invariant Solutions of Gas Dynamics: Proof Outline

Let $t \to t + t_0$ **where**

$$\begin{aligned} b &= \delta\tau \\ V &= \delta\tilde{V} \\ R &= \delta^s\tilde{R} \\ P &= \delta^{s+2}\tilde{P} \end{aligned} \tag{1.3}$$

Invariant Solutions of Gas Dynamics: Proof Outline

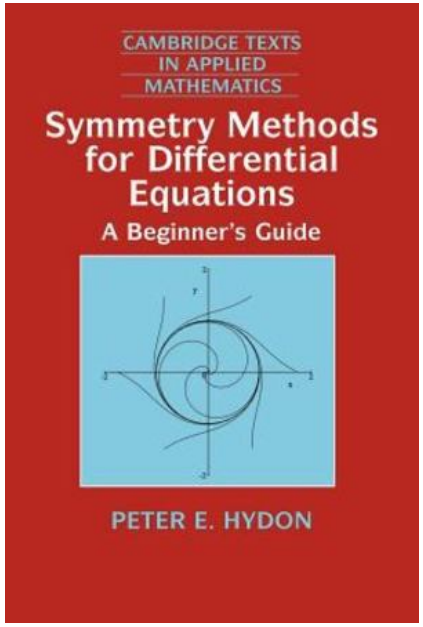
$$
\begin{aligned}
v &= \frac{r}{t}\tilde{V}(\lambda),\\
\rho &= \frac{a}{r^{k+3}t^{s}}\tilde{R}(\lambda),\\
p &= \frac{a}{r^{k+1}t^{s+2}}\tilde{P}(\lambda),\\
\lambda &= \frac{r}{r_0}e^{-t/\tau}
\end{aligned}
\tag{1.4}
$$

Very much like (1.2)

Invariant Solutions of Gas Dynamics: Comparison

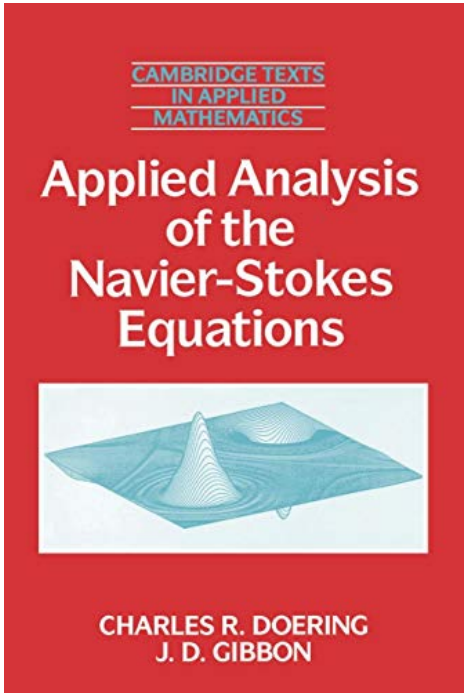
	Sedov (1.2)	**Proof** (1.4)
v	$\frac{r}{t}V(\lambda)$	$\frac{r}{t}\tilde{V}(\lambda)$
ρ	$\frac{a}{r^{k+3}t^{s}}R(\lambda)$	$\frac{a}{r^{k+3}t^{s}}\tilde{R}(\lambda)$
p	$\frac{a}{r^{k+1}t^{s+2}}P(\lambda)$	$\frac{a}{r^{k+1}t^{s+2}}\tilde{P}(\lambda)$
λ	$\frac{r}{bt^{d}}$	$\frac{r}{r_0}e^{-t/\tau}$

Symmetry Methods for Differential Equations



Hydon 2000

Applied Analysis of the Navier-Stokes Equations



Doering and Gibbon 1995

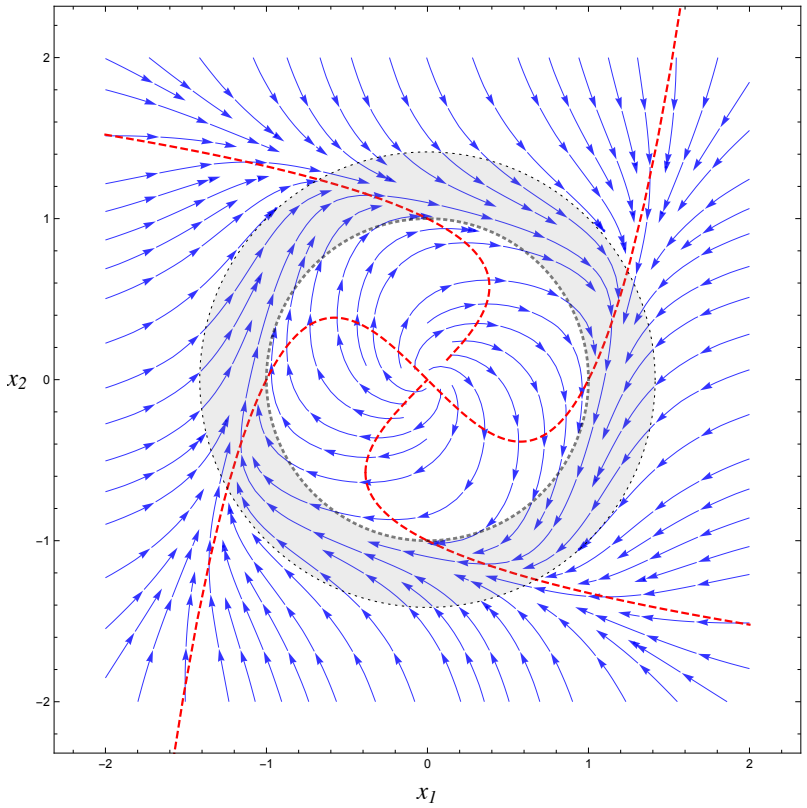
Temporal Derivative

$$\left(\frac{df(\mathbf{x},t)}{dt}\right)_{\textbf{fixed } \mathbf{x}} = \lim_{\delta t \to 0} \frac{f(\mathbf{x}, t+\delta t) - f(\mathbf{x},t)}{\delta t}$$

Temporal Derivative

$$\left(\frac{df(\mathbf{x},t)}{dt}\right)_{\textbf{fixed } \mathbf{x}} = \lim_{\delta t \to 0} \frac{f(\mathbf{x},t+\delta t) - f(\mathbf{x},t)}{\delta t} = \frac{\partial f(\mathbf{x},t)}{\partial t} \tag{3.1}$$

Go With The Flow



Convective Derivative: Definition

$$\left(\frac{df(\mathbf{x}+\mathbf{u}\delta t, t+\delta t)}{dt}\right)_{\textbf{stream } \mathbf{u}} = \lim_{\delta t \to 0} \frac{f(\mathbf{x}+\mathbf{u}\delta t, t+\delta t) - f(\mathbf{x}, t)}{\delta t} \quad (3.2)$$

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$$\left(\frac{df(\mathbf{x}+\mathbf{u}\delta t, t+\delta t)}{dt}\right)_{\textbf{stream } \mathbf{u}} = \lim_{\delta t \to 0} \frac{f(\mathbf{x}+\mathbf{u}\delta t, t+\delta t) - f(\mathbf{x}, t)}{\delta t} \quad (3.2)$$

Convective Derivative: Resolution

Method: Taylor Approximation Theorem

$$
\begin{aligned}
f(\mathbf{x}+\mathbf{u}\delta t, t+\delta t) &= f\left(x_1, t+\delta t\right) + u_1 \delta t f_1\left(\mathbf{x}, t+\delta t\right) \\
&+ f\left(x_2, t+\delta t\right) + u_2 \delta t f_2\left(\mathbf{x}, t+\delta t\right) \\
&+ f\left(x_3, t+\delta t\right) + u_3 \delta t f_3\left(\mathbf{x}, t+\delta t\right) + \mathcal{O}\left(\delta t^2\right) \qquad (3.3) \\
&= f\left(\mathbf{x}, t+\delta t\right) + \delta t\, \mathbf{u} \cdot \nabla f\left(\mathbf{x}, t+\delta t\right) + \mathcal{O}\left(\delta t^2\right)
\end{aligned}
$$

Convective Derivative: Resolution

$$\lim_{\delta t \to 0} \frac{f(\mathbf{x} + \mathbf{u}\delta t, t + \delta t)}{\delta t} = \lim_{\delta t \to 0} \frac{f(\mathbf{x}, t + \delta t)}{\delta t} + \cancel{\delta t} \frac{\mathbf{u} \cdot \nabla f(\mathbf{x}, t + \delta t)}{\cancel{\delta t}}$$
$$= \lim_{\delta t \to 0} \frac{f(\mathbf{x}, t + \delta t)}{\delta t} + \mathbf{u} \cdot \nabla f(\mathbf{x}, t) \tag{3.4}$$

Convective Derivative

$$\left(\frac{df(\mathbf{x}+\mathbf{u}\delta t, t+\delta t)}{dt}\right)_{\textbf{stream } \mathbf{u}} = \frac{\partial f(\mathbf{x},t)}{\partial t} + \mathbf{u}\cdot\nabla f(\mathbf{x},t) \tag{3.2}$$

Total Derivative

$$\frac{D}{Dt} = \partial_t + v^k \partial_k = \partial_t + \mathbf{v} \cdot \nabla \tag{3.5}$$

a.k.a convective derivative, material derivative

Material Derivative Also Know As...

1. **advective derivative (Majda 2003)**
2. **convective derivative**[1]
3. **derivative following the motion (Bird 2002)**
4. **hydrodynamic derivative (Bird 2002)**
5. **Lagrangian derivative**[2]

[1] Waves and compressible flow, Ockendon, H. and Ockendon, J. R., Springer, 2004.
[2] Introduction to Physical Oceanography, Mellor, G. L., Springer, 1996

Material Derivative Also Know As...

6. particle derivative (Stoker 1992)
7. substantial derivative (Bird 2002)
8. substantive derivative (Granger 1995)
9. Stokes derivative (Granger 1995)
10. total derivative (Bird 2002, Landau 1959)
11. gauge covariant derivative (Pokorski 2000)

Convective Derivative: Application

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + (\nabla \times \mathbf{u}) \times \mathbf{u} + \nabla \left(\frac{1}{2} \mathbf{u}^2 \right) \tag{3.6}$$

Conservation: Mass

$$\frac{\partial}{\partial t}\delta M = 0 \tag{3.7}$$

Rate of Change: Volume

$$\frac{\partial}{\partial t}\delta V = \sum_{k=1}^{3} \frac{d\delta x_k}{dt}\delta x_{k+1}\delta x_{k+2} \tag{3.8}$$

Using cyclical increments $k = 1, 2, 3$

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